

UNIVERSIDAD DISTRITAL FRANCISCO JOSÉ DE CALDAS

Facultad de Ingeniería



**UNIVERSIDAD DISTRITAL
FRANCISCO JOSÉ DE CALDAS**
Acreditación Institucional de Alta Calidad

Workshop 2

Advanced Learning Models (Part A): Deep Learning with Convolutional Neural Networks

Jhojan Stiven Aragón Ramírez - 20221020060

Santiago Reyes Gómez - 20221020068

Juan Andrés Jiménez Palomino - 20221020087

System Engineering

Professor: Carlos Andres Sierra

September 2026

Contents

1	Introduction	2
2	Architecture Selection	2
2.1	Problem characteristics	2
2.2	Alternatives considered	2
2.3	Selected approach	3
3	Model Description	3
3.1	SmallCNN — architecture trained from scratch	3
3.2	ResNet-18 transfer variant	4
4	Training Protocol	4
4.1	Data preparation	4
4.2	Optimization and regularization	5
4.3	Implementation and reproducibility	5
5	Training Results	6
5.1	SmallCNN (trained from scratch)	6
5.2	ResNet-18 (ImageNet transfer)	7
6	Test-Set Evaluation	7
6.1	Comparison with Workshop 1	9
7	Analysis	10
7.1	Grad-CAM on the SmallCNN	10
7.2	Grad-CAM on the transfer model	11
7.3	Consistency with the exploratory analysis	11
7.4	Limitations	12
8	Conclusions and Future Work	12

1 Introduction

Workshop 1 characterized the Fashion-MNIST dataset Xiao et al., 2017 in depth and established a set of traditional supervised learning baselines for the image classification task. The exploratory analysis confirmed optimal data integrity, perfect class balance ($IR = 1.00$), a strong concentration of discriminative information in the central region of the images, and a systematic overlap between the upper-body garment classes (T-shirt/top, Shirt, Pullover and Coat) that every classifier is expected to confuse. The best Workshop 1 baseline, a Histogram Gradient Boosting classifier, reached 87.48% test accuracy with a macro F1-score of 0.8740.

This second workshop addresses the same classification problem with deep learning. The goal is to design, train, evaluate and save a convolutional neural network that clearly outperforms the Workshop 1 baselines and produces meaningful learned representations. The contributions of this report are: (i) the selection and justification of the architecture against suitable alternatives; (ii) the description and implementation of the model in PyTorch; (iii) a training protocol aligned with the deep learning literature; (iv) the training results with learning curves; (v) the test-set evaluation of both models plus a qualitative analysis of the learned behavior through Grad-CAM visualizations; and (vi) the selection and verification of the final saved checkpoint.

The report includes the complete comparative analysis against the full set of Workshop 1 baselines, the results of the transfer-learning variant and the final model selection according to validation performance.

2 Architecture Selection

2.1 Problem characteristics

The task is a 10-class classification problem over 28×28 grayscale images. The discriminating information is spatial and localized: the Workshop 1 analysis showed that the image borders carry almost no information and that the signal is concentrated in the central region of the garment. Architectures that exploit spatial locality, weight sharing and translation invariance are therefore the natural candidates, which places convolutional neural networks (CNNs) at the center of the selection.

2.2 Alternatives considered

- **Transfer learning from ImageNet CNNs.** Architectures such as ResNet-18 He et al., 2015a, VGG Simonyan and Zisserman, 2014, Inception Szegedy et al., 2014 and EfficientNet Tan and Le, 2019 provide pretrained features that transfer well across vision tasks Yosinski et al., 2014; Zhuang et al., 2019, and the workshop

guidelines explicitly suggest leveraging them. The main drawback for this dataset is the domain gap: the pretrained models expect 224×224 color natural images, whereas Fashion-MNIST provides 28×28 grayscale silhouettes, so the input must be resized and channel-replicated before use.

- **Vision Transformers (ViTs).** Recent reviews compare CNNs and ViTs on Fashion-MNIST Bbouzidi et al., 2024; transformers are data-hungry and rely on large-scale pretraining or heavy augmentation, which makes them a less suitable fit for small grayscale inputs at this scale.
- **Compact CNN trained from scratch.** A small VGG-style network with 3×3 convolutions, batch normalization Ioffe and Szegedy, 2015 and a global-average-pooling classifier fits the native input resolution, is inexpensive to train and iterate on, and the available training data (49,000 images) is sufficient to train it without overfitting when regularized properly.

2.3 Selected approach

The primary model is a **compact CNN trained from scratch** (`SmallCNN`), designed specifically for 28×28 grayscale inputs. As a secondary track, a **ResNet-18 transfer variant** is trained with the two-stage fine-tuning procedure suggested by the workshop guidelines: first freeze the pretrained backbone and train only the classifier head, then unfreeze the upper layers and continue training with a lower learning rate. The transferability evidence justifies this schedule: general features in early layers transfer across tasks, while deeper features become task-specific Yosinski et al., 2014. Both tracks were implemented and trained; the final model was selected according to the validation macro F1-score (Section 5).

3 Model Description

3.1 SmallCNN — architecture trained from scratch

The network follows a VGG-style design: homogeneous 3×3 convolutions Simonyan and Zisserman, 2014, each followed by batch normalization Ioffe and Szegedy, 2015 and ReLU activation, with 2×2 max-pooling after each block and a global-average-pooling classifier head.

Stage	Output shape	Parameters
Input	$1 \times 28 \times 28$	—
Block 1: Conv 3×3 ($1 \rightarrow 32$) + BN + ReLU; Conv 3×3 ($32 \rightarrow 32$) + BN + ReLU; MaxPool	$32 \times 14 \times 14$	9,632
Block 2: Conv 3×3 ($32 \rightarrow 64$) + BN + ReLU; Conv 3×3 ($64 \rightarrow 64$) + BN + ReLU; MaxPool	$64 \times 7 \times 7$	55,552
Block 3: Conv 3×3 ($64 \rightarrow 128$) + BN + ReLU; MaxPool	$128 \times 3 \times 3$	73,984
Head: GlobalAvgPool $\rightarrow$ Dropout (0.5) $\rightarrow$ Linear ($128 \rightarrow 10$)	10	1,290
Total		140,458

Table 1: SmallCNN architecture. Output shapes use 28×28 inputs.

Weights are initialized with the He/Kaiming scheme for rectifier networks He et al., 2015b and the batch-normalization parameters with $\gamma = 1$, $\beta = 0$. A forward pass over a sample batch was verified during implementation: the model receives a tensor of shape $(B, 1, 28, 28)$ and produces logits of shape $(B, 10)$.

3.2 ResNet-18 transfer variant

The second track uses the torchvision implementation of ResNet-18 He et al., 2015a with ImageNet-pretrained weights. The original classifier is replaced by a linear layer $512 \rightarrow 10$, leaving 11,181,642 trainable parameters. Inputs are adapted to the pre-trained domain by replicating the grayscale channel to RGB, resizing to 224×224 and normalizing with ImageNet statistics. Fine-tuning follows the two-stage protocol: the backbone is frozen for the first 5 epochs (classifier-only training) and then unfrozen for the remaining epochs with a lower learning rate.

4 Training Protocol

4.1 Data preparation

The experiments reuse the exact data protocol of Workshop 1 to keep the results comparable: a stratified 70/15/15 split of the 70,000 images (49,000 train, 10,500 validation, 10,500 test) with a fixed random seed (42), and the global normalization statistics estimated in the Workshop 1 analysis ($\mu = 0.2860$, $\sigma = 0.3530$). Training images are augmented with random horizontal flips ($p = 0.5$) and random translations ($\pm 10\%$), mirroring the Workshop 1 augmentation; mixup Zhang et al., 2017 is available as an optional regularization and left for the ablation study.

4.2 Optimization and regularization

Component	Configuration
Loss function	Cross-entropy (label smoothing optional)
Optimizer	Adam, $\text{lr} = 10^{-3}$ (SGD + momentum as alternative) Kingma and Ba, 2014
Learning-rate schedule	Cosine annealing over the full run Loshchilov and Hutter, 2016
Weight decay	10^{-4}
Regularization	Batch Normalization Ioffe and Szegedy, 2015 + Dropout 0.5 in the head Hinton et al., 2012
Initialization	He/Kaiming for ReLU networks He et al., 2015b
Batch size / epochs	128 / 30
Early stopping	Patience 8 epochs on validation macro F1-score
Random seed	42

Table 2: Training configuration for the SmallCNN run.

Component	Configuration
Input	$224 \times 224 \times 3$ (grayscale replicated to RGB), ImageNet normalization
Pretrained weights	ImageNet (torchvision ResNet-18)
Optimizer	Adam, $\text{lr} = 3 \times 10^{-4}$ Kingma and Ba, 2014
Learning-rate schedule	Cosine annealing over the full run Loshchilov and Hutter, 2016
Weight decay	10^{-4}
Batch size / epochs	32 / 20 (5 epochs frozen backbone + 15 fine-tuning epochs)
Early stopping	Patience 8 epochs on validation macro F1-score (not triggered)
Random seed	42

Table 3: Training configuration for the ResNet-18 transfer run.

4.3 Implementation and reproducibility

Models are implemented in PyTorch. The SmallCNN was trained on Apple Silicon (MPS backend) at roughly 23 seconds per epoch; the ResNet-18 run was executed on an AMD Radeon RX 6600 (4 GB) through the DirectML backend in about 93 minutes (≈ 5.6 minutes per epoch during fine-tuning) using batch size 32 to fit the GPU memory, and both scripts select the available device automatically (CUDA, MPS, DirectML or CPU). Training and validation loss, accuracy and macro F1-score are logged every epoch to a CSV file and to TensorBoard, and learning curves are generated from those logs. The best-performing checkpoint (highest validation macro F1-score) is saved to-

gether with the class list, normalization statistics and the full training configuration, so that the evaluation can be reproduced exactly with a single command.

5 Training Results

5.1 SmallCNN (trained from scratch)

The SmallCNN was trained for 30 epochs with the protocol of Table 2. Figure 1 shows the learning curves: training and validation loss decrease smoothly and monotonically, and both accuracy and macro F1-score follow the same behavior without any sign of divergence between the curves.

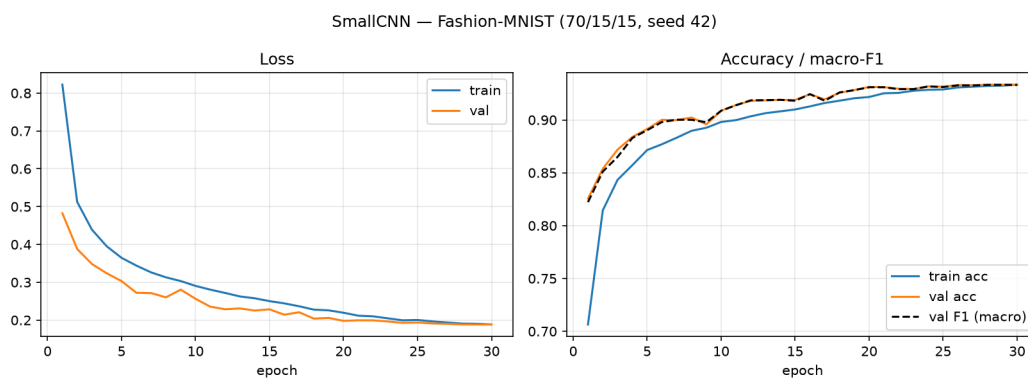


Figure 1: Learning curves of the SmallCNN on the 70/15/15 split (seed 42): loss (left) and accuracy / macro F1 (right).

The model improved from 82.6% validation accuracy after the first epoch to a best value of 93.33% validation accuracy with macro F1-score 0.9332 at epoch 29. Two observations stand out:

- **No overfitting.** The final training accuracy (93.34%) and validation accuracy (93.32%) are practically identical, and the training loss keeps decreasing: the effect of dropout, batch normalization and augmentation is sufficient at this model capacity.
- **The schedule, not the capacity, was the limiting factor.** Both losses were still decreasing when the cosine schedule reached its end; longer runs or larger models would likely improve the result further.

5.2 ResNet-18 (ImageNet transfer)

The transfer variant followed the two-stage schedule of Table 3. During the first five epochs the pretrained backbone stayed frozen and only the new classifier was trained; the model reached 83.30% validation accuracy with macro F1-score 0.8342 at the end of this stage. Once the backbone was unfrozen and the learning rate lowered to 3×10^{-4} , performance jumped to 90.66% at epoch 6 and improved steadily until the best values of the run: 95.53% validation accuracy and macro F1-score 0.9553 at epoch 20, the last epoch of the cosine schedule (Figure 2).

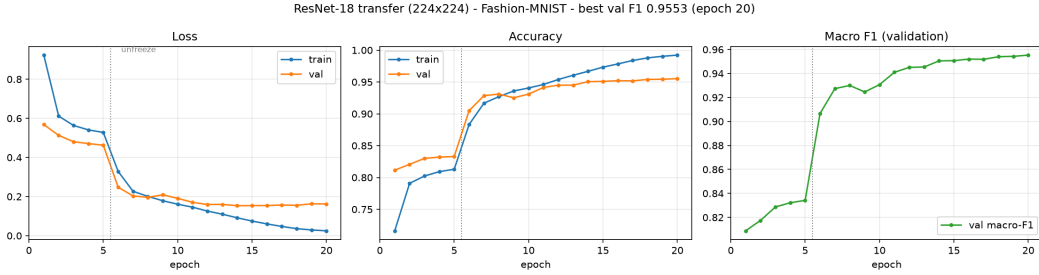


Figure 2: Learning curves of the ResNet-18 transfer run (seed 42). The dotted line marks the transition between the frozen stage (epochs 1–5) and fine-tuning (epochs 6–20).

The run took 93.4 minutes on an AMD Radeon RX 6600 (4 GB) through the DirectML backend, using a reduced batch size of 32 to fit the available GPU memory (≈ 1.5 min per epoch while the backbone was frozen and ≈ 5.6 min per epoch during fine-tuning; the same configuration is estimated at ≈ 17 hours on CPU). As with the from-scratch model, both losses were still decreasing when the schedule ended, so the best checkpoint is the last epoch and further gains are expected from longer schedules.

6 Test-Set Evaluation

The best checkpoints of both models were evaluated on the held-out test split of 10,500 images, using the same metrics as Workshop 1. Table 4 reports the aggregate results and Figures 3 and 4 show the corresponding confusion matrices.

Metric	SmallCNN	ResNet-18
Accuracy	93.05%	94.94%
Precision (macro)	0.9302	0.9494
Recall (macro)	0.9305	0.9494
F1-score (macro)	0.9302	0.9494
ROC-AUC (OvR, macro)	0.9961	0.9979

Table 4: Test-set metrics of the two models ($n = 10,500$).

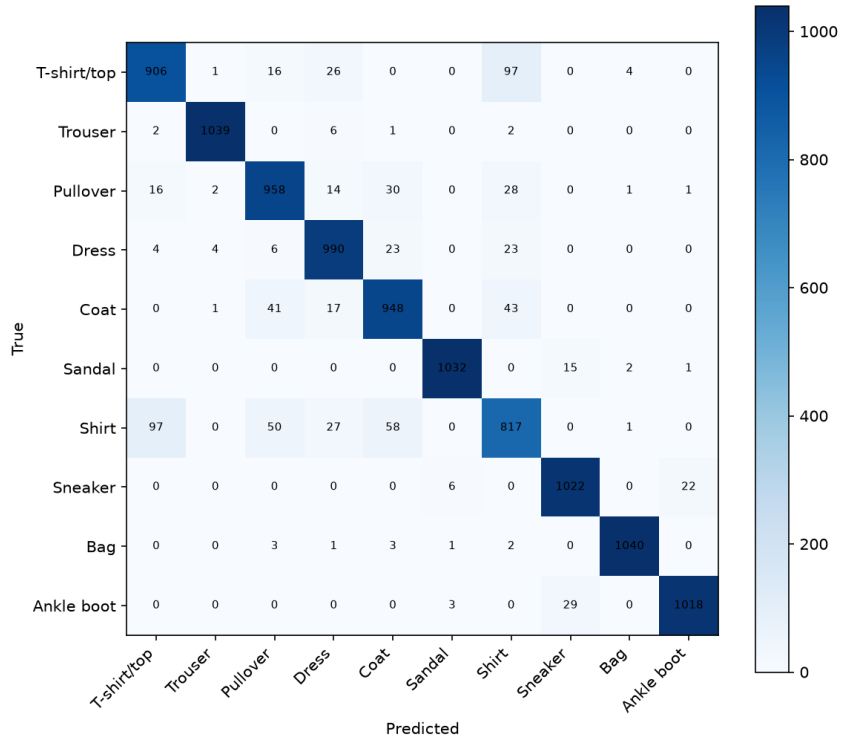


Figure 3: Confusion matrix of the SmallCNN on the test split.

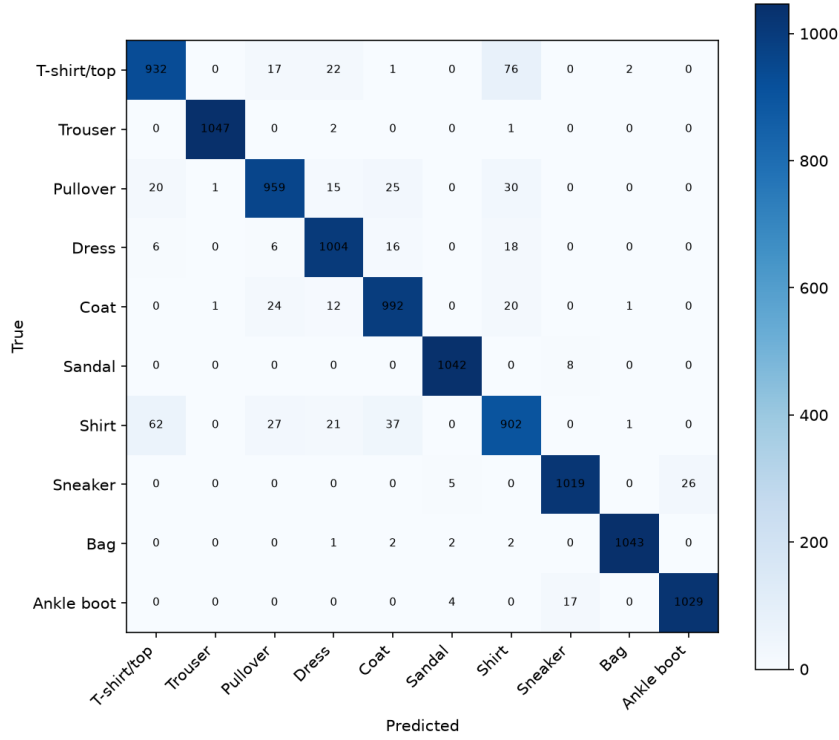


Figure 4: Confusion matrix of the ResNet-18 transfer model on the test split.

Both models share the same error structure, which confirms the analysis anticipated in Workshop 1. Footwear and accessories are classified almost perfectly by the transfer model (F1-scores: Trouser 0.998, Bag 0.995, Sandal 0.991, Ankle boot 0.978, Sneaker 0.973), while the upper-body garments concentrate the errors. The transfer model improves every difficult class with respect to the from-scratch network: **Shirt** goes from F1 0.79 to 0.86 and T-shirt/top from 0.87 to 0.90, with the residual confusions still spread among Shirt, Pullover, Coat and T-shirt/top.

6.1 Comparison with Workshop 1

Table 5 places both deep models next to the full set of Workshop 1 baselines, all evaluated on the same test split. This completes the comparative analysis that the previous revision of this report had reserved.

Model	Test accuracy	Macro F1
Logistic Regression (W1)	85.31%	0.8519
Random Forest (W1)	85.90%	0.8565
HistGradientBoosting (W1, best baseline)	87.48%	0.8740
SmallCNN (W2, from scratch)	93.05%	0.9302
ResNet-18 (W2, transfer)	94.94%	0.9494
Difference (best W2 – best W1)	+7.46 pts	+7.54 pts

Table 5: Full test-set comparison: Workshop 1 baselines versus the two Workshop 2 deep models, on the same 70/15/15 split (seed 42).

The transfer model outperforms the strongest traditional baseline by 7.5 points of accuracy and macro F1, and improves over the from-scratch CNN by 1.9 points. Both deep models satisfy the workshop expectation that the learned models clearly improve over the Workshop 1 results.

7 Analysis

7.1 Grad-CAM on the SmallCNN

To inspect the learned behavior, Grad-CAM Selvaraju et al., 2016 was applied to the trained network: for a given prediction, the method weights the last convolutional feature maps by the gradient of the predicted class score, producing a coarse map of the image regions that support the decision. Figure 5 shows three correct classifications (top row) and the three most confident misclassifications (bottom row).

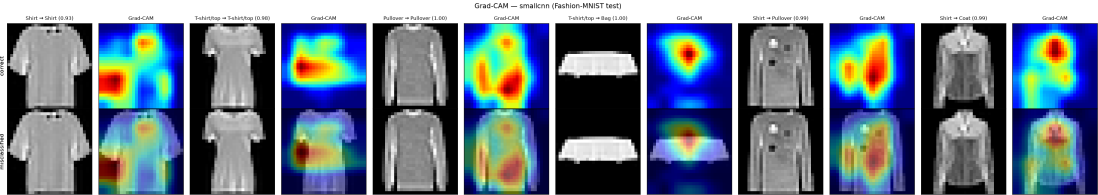


Figure 5: Grad-CAM examples. Each pair shows the input image and its class-activation map; titles read *true* → *predicted* (confidence).

Three observations follow from the visualizations:

- **Attention is placed on the garment, not the background.** For correct predictions the activation concentrates on the body of the garment (torso, chest and

sleeves), which is evidence that the network learned domain-relevant features rather than exploiting the constant image borders.

- **Upper-body confusions come from shared texture and silhouette.** The Shirt $\rightarrow$ Pullover and Shirt $\rightarrow$ Coat errors activate regions dominated by knit texture and sleeve shape, i.e., features shared across the upper-body classes, instead of the collar and cut details that actually separate them.
- **Confident errors involve genuinely ambiguous samples.** The most confident error (T-shirt/top $\rightarrow$ Bag, 0.997) corresponds to an unusual cropped top whose silhouette overlaps with the bag cluster. This mirrors the Workshop 1 finding that footwear and bags form compact clusters: atypical upper-body items can fall inside them.

7.2 Grad-CAM on the transfer model

Figure 6 repeats the analysis for the ResNet-18. The attention maps confirm the same behavior at a finer spatial scale: correct predictions (top row) concentrate on the torso and chest of the garment, while the three most confident errors (bottom row) are again upper-body confusions — T-shirt/top $\rightarrow$ Pullover, Shirt $\rightarrow$ T-shirt/top and T-shirt/top $\rightarrow$ Shirt, all with probability 1.00 — whose activation covers the whole torso, exactly the region shared by the four overlapping classes. The transfer model reduces the error rate of these classes (Shirt F1 0.79 $\rightarrow$ 0.86, T-shirt/top 0.87 $\rightarrow$ 0.90) but the samples shown remain genuinely ambiguous silhouettes.

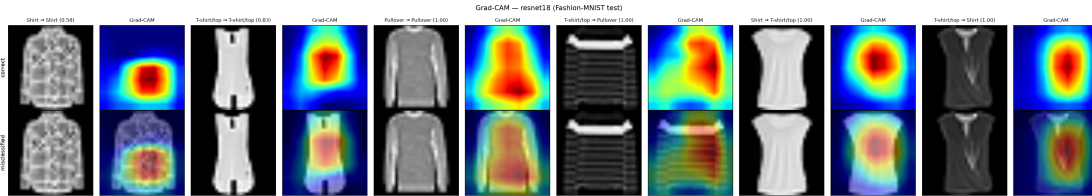


Figure 6: Grad-CAM examples for the ResNet-18: three correct classifications (top row) and the three most confident errors (bottom row). Titles read *true* $\rightarrow$ *predicted* (confidence).

7.3 Consistency with the exploratory analysis

The error structure observed on the test set matches the predictions made in Workshop 1 with the PCA projection and the mean images: footwear and bags are well separated ($F1 \geq 0.97$), while T-shirt/top, Shirt, Pullover and Coat overlap and absorb most of the

residual error. The deep model improves the general performance substantially, but it does not (and should not be expected to) remove the intrinsic ambiguity of those four classes.

7.4 Limitations

- The reported results come from a single training run with a fixed seed; the sensitivity to the seed is not yet characterized.
- The transfer model adapts the input by replicating the grayscale channel to RGB and upsampling to 224×224 ; lighter native-resolution transfer setups (e.g., 64×64 inputs) remain as future work.
- Hyperparameter choices follow the literature and the workshop guidelines; a systematic ablation study (mixup, optimizer, schedule, dropout rate) is left for future work.

8 Conclusions and Future Work

This workshop implemented, trained, evaluated and saved two deep convolutional models for Fashion-MNIST image classification. The compact CNN trained from scratch (140,458 parameters) reached 93.05% test accuracy with macro F1-score 0.9302, while the ImageNet-pretrained ResNet-18, fine-tuned in two stages (11.2M parameters), reached 94.94% and 0.9494. Both clearly outperform the strongest Workshop 1 baseline (HistGradientBoosting, 87.48% accuracy and 0.8740 macro F1) by more than five points, and the full test-set comparison against all three traditional baselines is included in this report. Following the workshop’s selection rule (validation macro F1-score), the ResNet-18 transfer model is the final model: its best checkpoint (epoch 20, validation macro F1-score 0.9553) is saved for reuse. The learning curves show no overfitting in either run, and the Grad-CAM analysis confirms that both models base their decisions on the garment itself, with the residual errors concentrated in the intrinsically overlapping upper-body garment classes — exactly as the Workshop 1 exploratory analysis anticipated.

The following work remains open:

- Characterize the sensitivity of the results to the random seed and to the data split.
- Run ablation experiments on the optional regularizers (mixup, label smoothing), on the optimizer choice (Adam vs. SGD with momentum) and on the input resolution of the transfer model.

- Extend the Grad-CAM analysis to a per-class review, including the correctly classified versus confused examples of the upper-body garments.

References

- Bbouzidi, S., Hcini, G., Jdey, I., & Drira, F. (2024). Convolutional neural networks and vision transformers for fashion mnist classification: A literature review. <https://arxiv.org/abs/2406.03478>
- He, K., Zhang, X., Ren, S., & Sun, J. (2015a). Deep residual learning for image recognition. <https://arxiv.org/abs/1512.03385>
- He, K., Zhang, X., Ren, S., & Sun, J. (2015b). Delving deep into rectifiers: Surpassing human-level performance on imagenet classification. <https://arxiv.org/abs/1502.01852>
- Hinton, G. E., Srivastava, N., Krizhevsky, A., Sutskever, I., & Salakhutdinov, R. R. (2012). Improving neural networks by preventing co-adaptation of feature detectors. <https://arxiv.org/abs/1207.0580>
- Ioffe, S., & Szegedy, C. (2015). Batch normalization: Accelerating deep network training by reducing internal covariate shift. <https://arxiv.org/abs/1502.03167>
- Kingma, D. P., & Ba, J. (2014). Adam: A method for stochastic optimization. <https://arxiv.org/abs/1412.6980>
- Loshchilov, I., & Hutter, F. (2016). Sgdr: Stochastic gradient descent with warm restarts. <https://arxiv.org/abs/1608.03983>
- Selvaraju, R. R., Cogswell, M., Das, A., Vedantam, R., Parikh, D., & Batra, D. (2016). Grad-cam: Visual explanations from deep networks via gradient-based localization. <https://arxiv.org/abs/1610.02391>
- Simonyan, K., & Zisserman, A. (2014). Very deep convolutional networks for large-scale image recognition. <https://arxiv.org/abs/1409.1556>
- Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., Erhan, D., Vanhoucke, V., & Rabinovich, A. (2014). Going deeper with convolutions. <https://arxiv.org/abs/1409.4842>
- Tan, M., & Le, Q. V. (2019). Efficientnet: Rethinking model scaling for convolutional neural networks. <https://arxiv.org/abs/1905.11946>
- Xiao, H., Rasul, K., & Vollgraf, R. (2017). Fashion-mnist: A novel image dataset for benchmarking machine learning algorithms. <https://arxiv.org/abs/1708.07747>
- Yosinski, J., Clune, J., Bengio, Y., & Lipson, H. (2014). How transferable are features in deep neural networks? <https://arxiv.org/abs/1411.1792>
- Zhang, H., Cisse, M., Dauphin, Y. N., & Lopez-Paz, D. (2017). Mixup: Beyond empirical risk minimization. <https://arxiv.org/abs/1710.09412>
- Zhuang, F., Qi, Z., Duan, K., Xi, D., Zhu, Y., Zhu, H., Xiong, H., & He, Q. (2019). A comprehensive survey on transfer learning. <https://arxiv.org/abs/1911.02685>